An automatic approach to map refreezing ice in radar sounder data

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ABSTRACT

Radar sounders mounted on airborne platforms have acquired data on the subsurface of the Earth's icy areas over the last decades. These data, called radargrams, contain information on the dielectric discontinuities in the ice sheets, and thus on the buried geological structures and the related processes. Conventionally, these structures have been characterized and mapped by visually inspecting the radargrams. However, visual inspection is subjective and time-consuming and can lead to misinterpretations. Recently, state-of-the-art automatic techniques are proposed to map the position of the bedrock, the ice layering, and the noise in the radargram. However, there are no automatic techniques for mapping the basal refreezing, which is an important ice target that controls the rate of sea-ward flow of the ice sheets. This paper proposes an automatic method to map the refreezing ice in radargrams. We model the refreezing ice considering its geophysical and radiometric properties. Then, we design a set of features considering this model to perform a classification of the radargrams into four classes, i.e., ice layering, echo-free zone (EFZ) and thermal noise, bedrock, and the refreezing ice. We applied the proposed method to radargrams acquired in north Greenland by a Multichannel Coherent Radar Depth Sounder (MCoRDS3), a radar sounder designed by the Center for Remote Sensing of Ice Sheets (CReSIS). The results indicate a good overall accuracy. The accuracy of refreezing ice is high, while that of the other classes is comparable with state-of-the-art techniques. The results indicate the effectiveness of the proposed features in mapping the refreezing ice.

Keywords: Refreezing Ice, Ice Sheet, Automatic Classification, Radar sounder, Remote sensing

1. INTRODUCTION

Ice sheets feed ice shelves that calve into the sea, leading to the global sea-level rise. The see-ward flow of the ice depends on the condition at the basal interface and the quantity of water lubricating the ice-bedrock interface. In basal melting, a film of water forms above the bedrock which increases the sliding movement of the glacier towards the coastal regions. On the contrary, in basal freezing, the ice is frozen to the bedrock and thus hinders its flow. These conditions depend on the glacier thermal regime, the bedrock topography and deformation, the subglacial hydrology (e.g., subglacial lakes), and the melting-refreezing process 1.

The presence of water at the interface, e.g., subglacial lakes, ice flows, and water films, is strongly related to the stability and evolution of the glaciers 2, 3. Carter et al.[4] pointed out that the water at the rock/ice interface creates particular structures when it comes in contact with the ice. This structure, called refreezing ice, is located above the bedrock and elongates for several kilometers. Refreezing ice can extend up to half of the ice sheet thickness and causes deflection of the ice layering toward the surface. Accurate modeling of the basal processes, i.e., basal melting and refreezing, requires the direct measurements of the ice sheet up to the bedrock.

The information on the ice-sheet structure is imaged in radargrams, i.e., the data acquired by radar sounders. Radar sounders are active sensors that transmit an electromagnetic wave at a low frequency and with a wide

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bandwidth. These waves interact with the buried geological structures, and a part of the wave is reflected back to the antenna from each dielectric discontinuities. The coherent sum of these reflections is collected in radargrams. Thus, radargrams contain information on the buried geological structures in the ice sheet, e.g., ice layering, refreezing ice, bedrock.

Airborne radar sounders have been profiling the polar regions of the Antarctica and Greenland for decades. A huge amount of data is available from these acquisitions and is expected to grow further in the future. From these data, we can extract geophysical information on the glacier structure, the topography of the bedrock, and the glacier hydrology. Currently, the main approach to extract information is by visually inspecting radargrams. Bell et al.[5,6] analyzed refreezing ice by visually inspecting radargrams to define the possible cooling mechanisms, the perturbation in the shape and reflected power of the surrounding ice (i.e., the upward deflection of the internal layers), and the relation with the surface and bedrock topography.

Although this approach supports the characterization of several targets, it has several limitations considering the increasing amount of available data. This strategy is time-demanding and thus is not suitable for analyzing a large number of radargrams. Moreover, visual inspection is a subjective task that can lead to inconsistency and misinterpretations. Hence, new methods are needed to address these limitations and to automatically extract information on subsurface geological targets and processes.

A few automatic techniques exist that extract information about the ice sheet from the radargrams, although none of them map the refreezing ice. Ilisei et al.[7] segmented radargrams, considering three targets: the layers of ice, the bedrock, and the echo-free zone and thermal noise. The method extracts a set of features based on the spatial and statistical properties of the targets and uses them to performs a classification of the radargram.

The identification of the type of the basal boundary condition in a region, has been supported by complex models by combining different types of data. Schäfer et al.[8] proposed a method to define the basal condition in the Vestfonna ice cap (Nordaustlandet, Svalbard island) that considers the maps of the ice temperature profile, the surface velocity, the ice flow model, and the topography of the bedrock. They extracted only the topographic information on the bedrock from the radargrams. Christianson et al.[9] investigated the basal condition in the Whillans Ice Stream in the West Antarctic to define different types of interfaces, e.g., in presence of an ice stream and of ice shelves, and with different levels of basal roughness. The approach analyzes the bedrock reflectivity and waveforms of the radargram, to define the position and the reflectivity of the groundline (i.e., the boundary between rock and seawater under ice shelves) and of the basal interface. These approaches identified only the depth of the bedrock from radargrams, while other studies 4, 6, 10 indicate that additional information can be extracted, including the presence of water at the interfaces, to better model the basal conditions.

Some automatic techniques analyze the disruptions in the ice layers above the refreezing but do not consider the interface between the bedrock and the ice. These techniques determine the layering by using a line detection algorithm and then analyzing the slopes of the lines extracted 11, 12. Sime et al.[11] proposed a method to obtain the dip angles of englacial reflectors. Panton et al.[12] mapped the layering structure of the GrIS (Greenland Ice Sheet) and analyzed the disturbance in the layer slopes in relation to both their depth and the sliding velocity of the glaciers. However, these techniques focus on the analysis of the ice stratigraphy disruption, without considering the reflections due to the refreezing ice.

This paper proposes an automatic method to map the refreezing ice in radar sounder data acquired in Greenland. The method aims at assigning labels to all the samples in the radargrams, including those of the refreezing ice. It consists of three steps: first, we define a geophysical and radiometric model indicating the properties of the reflections generated by refreezing ice in the radargram. Second, we define a set of features according to the geo-electrical model. These features estimate the radiometric, spatial, and statistical properties of the refreezing ice signal in the radargrams. Finally, we segment the radargram into four classes, i.e., (i) refreezing ice, (ii) ice layering, (iii) bedrock, and (iv) echo/free zone and thermal noise. The method is tested on MCoRDS radargrams acquired in Greenland that show a significant presence of refreezing ice. The results indicate the effectiveness of the method in segmenting the radargram and in locating refreezing ice. The paper is structured as follows – Section 2 explains the proposed method, and Section 3 describes the dataset and the experimental results. Finally in Section 4, we draw the conclusion of the paper.

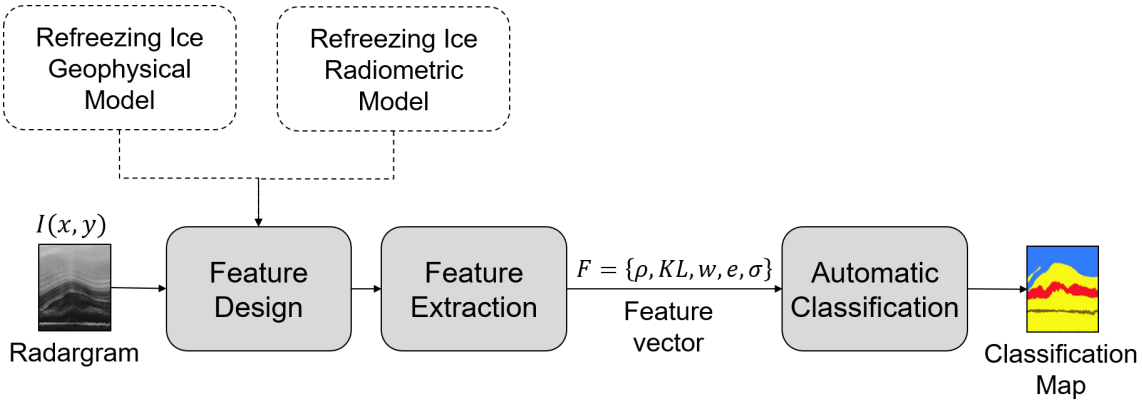


Figure 1: Block scheme of the propose method.

2. PROPOSED METHOD

Let us consider the radargram as a 2D matrix of n_T traces and n_S samples as follow

$$\mathbf{I} = \{I(x,y)|x \in X = [1,...,n_T], y \in Y = [1,...,n_S]\}$$

where I is the intensity of the reflections stored in the radargrams. The method focuses on the analysis of radargrams of icy areas to extract information on the basal boundary conditions, and more specifically on refreezing ice and melting. The aim is to locate the refreezing ice and consists of three steps, as shown in Figure 1:

1. define a model for the refreezing ice, considering its geological and radiometric properties, see Section 2.1;
2. design a set of features according to this model that discriminate refreezing ice from other structures, see Section 2.2;
3. perform the classification of radargrams taking in input the set of features in point 2 to segment the radargram in refreezing ice, layers, bedrock, and echo-free zone (EFZ) and thermal noise, see Section 2.3.

2.1 Model for Refreezing Ice

Figure 2 is an example of the radargrams acquired in Greenland and the Antarctica where we can see four main icy targets. Moving down along the range, on the top there are reflections generated by the ice strata. Next is the echo-free zone (EFZ) that represents a region which reflects waves having lower power than the sensitivity of the antenna. Below the EFZ, there is the reflection from the bedrock at the bottom of the ice sheet. Finally, the last part of the radargram is characterized by the thermal noise, since no echoes are reflected from this region.

Refreezing-Ice Geophysical Model. refreezing ice (shown as freeze-on ice) occurs at the base of the ice sheet in the presence of melt-water at the basal interface, i.e., the ice-bedrock interface 4, 13 (see Fig 2). Close to the core of the refreezing ice, the layers fold toward the surface, generating structures with really steep edges, called disrupted layers 6, 12. The upper part of the layers, close to the surface, are usually not affected by the refreezing, and thus, they remain flat. Under the freeze-on ice, there is the echo-free zone (EFZ) 7, where the signal reflected by the layers is lower than the sensitivity of the radar's antenna, and thus nothing is detected. Finally, the last bright reflection is due to the upper part of the basal interface. Note that in the presence of refreezing, the shape of the bedrock should be different from the shape of the disrupted layers.

Refreezing-Ice Radiometric Model. Refreezing ice is characterized by strong reflections, having four main radiometric properties. First, the signal in the region of the folded layers, above the freeze-on ice core, is characterized by fading due to the redistribution of the energy in the spectrum 13. Secondly, the refreezing ice is characterized by diffuse scattering that can be modeled with the specularity in the along-track direction,

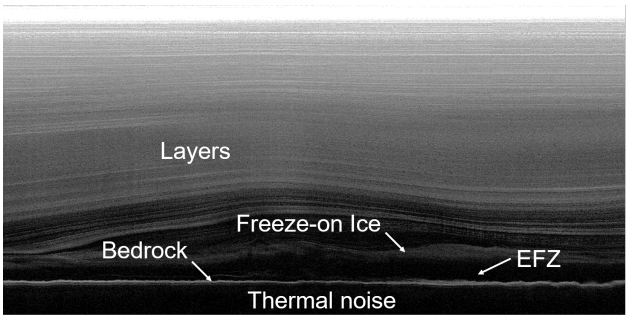


Figure 2: Detail of a refreezing area in a radargram acquired in North Greenland, CReSIS campaign 20170413_01.

Table 1: List of features derived based on the refreezing ice model

Model	Features
Spatial Distribution	Range Position
Statistical Distribution	Kullback-Leibler Divergence Map
Layer Disruption	Edge Operator
Energy Fading	Radar Equation
Diffuse Scattering	Specularity
Ice Temperature	Absolute attenuation coefficient

i.e., the amplitude of the echo strength with respect of the surrounding. The specularity inside the core of the freeze-on ice is greater than in the upper layers and the EFZ 4,10. Thirdly, the temperature of the refreezing ice is higher than the rest of the ice sheet, as it is newly generated and is subjected to the higher pressure by the ice sheet. Studies 4, 10 have analyzed the absolute reflection coefficient that can be calculated from the radar equation by considering the ice attenuation coefficient at the radar working frequency and the ice temperature. Young et al.[10] point out that a loss in the reflection power can be due to deformation of the layers, impurities (e.g., soil or stones), or a higher ice temperature. Note that the absolute reflection coefficient depends on the temperature of the ice, and thus it is complex to calculate. Finally, the last radiometric characteristic regards the bedrock: according to many studies 4–6, 10, in presence of refreezing ice, water is generally present at the interface between the ice and the bedrock as a thin film or even a lake. Hence, the bedrock under freeze-on ice should appear as an anisotropic specular reflector, i.e., it should produce a smooth and bright reflection.

2.2 Feature Design

This section describes the second step of the proposed method that aims at defining a set of features (see Table 1) based on the model defined in Section 2.1. We defined a set of features $F = [\rho_i, KL_i, w_i, e_i, \sigma_i]$, that describes the characteristics of the four classes.

Range Position. The feature ρ is the range position of the pixels with respect to the surface. It was first introduced by Ilisei et al.[7] and describes the spatial separation along the range direction of the target classes. This feature is defined as

$$\rho(x, y) = x - y_{max}(y), \forall x \in X = [1, ..., n_T], \forall y \in y = [1, ..., n_S])$$

where y_{max} is the position of the surface that can be analytically defined as the maximum return power for each trace y, and thus $y_{max} = max\{I(x, y)\}, x \in X = [1, ..., n_T], y \in y = [1, ..., n_S]$. The model in Section 2.1 shows that the targets are clearly separated in the range direction in the following order: ice layering, refreezing ice, EFZ, bedrock and thermal noise. Thus, the feature ρ will assume lower values for the samples related to the ice layering, while it will have higher values for the samples related to the bedrock and thermal noise. The refreezing ice, being closer to the bedrock, will have intermediate values.

Kullback-Leibler divergence map. The feature KL indicates the Kullback-Leibler (KL) divergence between two different statistical distributions. This feature was first introduced by Ilisei et al.[7] to separate the pixels dominated by the noise, i.e., EFZ and thermal noise, from those dominated by the signal, i.e., layers, bedrock,

and refreezing ice. First, a region containing noise-dominant samples is selected from the radargram below the bedrock, where there are no reflections. The noise and the amplitude samples are fitted with the Gamma distribution to retrieve the shape a and scale b parameters. We choose the Gamma distribution since Ilisei et al.[7] showed that it better estimates the probability density function (pdf) of the different classes. The KL distance is computed as in (1), where $H(A)$ is the fitted Gamma distribution of the amplitude samples $A(x, y)$, and M is the fitted pdf of the noise previously estimated.

$$KL(H, M) = \sum_{A} H(A) \log \frac{H(A)}{M(A)} \tag{1}$$

The values of this feature are proportional to the difference between the two distributions. Hence, the more the distributions are different, as in the case of noise and signal, the more the feature will have higher values. On the contrary, when comparing similar distributions, as in the case of noise and noise, the resulting KL divergence will tend to zero.

Edge Detector. The feature w aims at identifying the layer disruption in the lower part of the ice stratigraphy and the refreezing ice. Compared to the upper part of the ice layering and the bedrock, these regions are characterized by highly oscillating reflections. Thus, the feature w is computed by applying an edge detector, such as the Sobel operator or the wavelet decomposition. Here, we apply the wavelet decomposition since it has been shown to perform well in both SAR and radar sounder data for layer detection and image denoising 14.

We apply a 2D orthogonal discrete wavelet transform (DWT) to decompose the radargram $\mathbf{I}$ into a set of images characterized by a lower level of noise and with enhanced layers. To this end, we decompose the radargram $\mathbf{I}$ in sub-bands following the Mallat pyramidal scheme. This is done by recursively applying independently two filters on the range and azimuth directions. These filters have a high $h(.)$ and a low $l(.)$ impulse response, respectively. Thus, after each step, the radargram is decomposed into a low-resolution image $\mathbf{I_{LL}}$, corresponding to lower frequencies in the azimuth and range directions, and into three detail-preserving images $\mathbf{I_{HH}}, \mathbf{I_{LH}}, \mathbf{I_{HL}}$ related to the higher frequencies in the range (HH direction) and diagonal directions (LH, HL directions).

In this work, we apply Daubechies wavelet since it proved to perform well on radar data. We decompose the original radargram with two levels of decomposition, and we choose the decompositions whose energy sum is higher than 99% of the energy of the original radargram $\mathbf{I}$. This eliminates the decomposed images containing noise, while preserving those with the details. Thus, feature w consists of a set of decomposed images $w = \{\mathbf{I^n_d}, n = [1, ..., N], d = [HH, HL, LH, HH]\}$, where n indicates the level of decomposition and d the type of decomposition. The flat reflections of the upper part of the ice stratigraphy will have a higher response to the horizontal wavelet decomposition. While those of the lower part, characterized by strong undulation, will respond better to the diagonal decompositions. Applying the wavelet transform to the power signal in dB mitigates the effects of the noise, which can be modeled as an additive model and not multiplicative.

Energy fading. The feature e is related to energy fading occurring in the lower regions of the ice stratigraphy, above the steep edges of the deflected layers. Oswald et al.[13] explains that the energy redistributes in the spectrum, and thus the wave reflected to the antenna has lower energy. This appears in radargrams as energy fading in the regions above refreezing ice.

The energy redistributed in the spectrum can be calculated as the difference between the energy in the radargram and that expected assuming a homogeneous ice-sheet. This feature is estimated as $e = P_r(x, y)/P_n(x, y)$, where $P_r(x, y)$ is the power in the radargram and $P_n(x, y)$ is the power expected with no redistribution. $P_n(x, y)$ is calculated from the radar equation considering a homogeneous value for the relative permittivity of the ice. In the case of energy fading due to refreezing ice, this feature has a high value. Instead, more the layers are flat, more this value will tend to zero.

Specularity. The feature σ models the specularity, i.e. the directional reflectivity of the ice-sheet structures. The specularity indicates the variation of the echo strength across the traces. Here, we calculate this variation by estimating the standard deviation of the radargram. The standard deviation is defined as $\sigma = \sqrt{E[(P_r(x, y) - E[P_r(x, y)])^2]}$, where $E[.]$ is the expectation operator, and $P_r(x, y)$ is the radargram power. This feature assumes higher values in the regions where the power of the pixels is highly varying, representing diffused scattering (such as in the case of the refreezing ice reflections). On the contrary, the value of this feature

Table 2: Parameter of the radar sounder instrument MCoRDS[15] and the data.

Parameter	Value
Central frequency f_c	195 MHz
Bandwidth BW	30 Mhz
Transmitted Power P_{tr}	1050 W
Aircraft Amplitude h	2-4 km
Range Resolution in Ice	4.3 m
Azimuth Resolution	25 m
Acquisition Campaign	20170413_01 in Greenland
Radargrams	45-46
Number of Traces	6000
Number of Samples	1143

tends to zero when the homogeneity of the neighboring pixels is high, representing specular scattering (such as from the bedrock or flat layers).

Features KL, σ are extracted from the radargram through a sliding window approach. The (n_w x n_h) window has a rectangular shape since the target classes, i.e., the layers, the bedrock, the refreezing ice, are elongated in the azimuth direction. Note that the dimensions, i.e., both the width n_w (along the azimuth direction) and the height n_h (along range direction), have a great importance since the window should be large enough to be representative of the data. At the same time, the dimensions should be small enough to avoid the low-pass filtering effect at the borders of radargrams.

2.3 Automatic Classification

This step aims at segmenting the radargram in four regions, i.e., ice layering, refreezing ice, bedrock, and thermal noise and echo-free zone. We apply a pixel-based classification that learns the characteristics of the different classes and assigns a label to each pixel. We select the SVM classifier since it is known to perform well in binary and multiclass classification tasks 7. Moreover, it has several advantages: it has a good generalization capability and avoids over-fitting; ii) it is able to solve non-linear problems in the feature space; and iii) the solution is sparse and unique. Anyway, once can use any other classifier.

A supervised classifier learns the characteristics of the different classes from a set of labeled samples, i.e. the training set. The training set T consists of several couples of samples and the associated labels assigned by an expert. In this case, the training set contains a subset of the total samples for each class, i.e., $T = \{v_j, q_j^c\}, j = [1, .., J_c], c = [1, ..., C]$, where v_j is the sample associated with label q_j^c, and c indicates the class. The number of samples for each class J_c is proportional to the area occupied by the class in the radargram. Each sample is described by the array of features defined in Section 2.2.

During the training phase, the classifier learns to identify the class corresponding to a given radargram sample. It calculates a decision function $g(v)$ that can discriminate between the different classes and assign a label to the unknown samples v. To improve the discrimination performances, generally a suitable mathematical transformation function is applied to the input data. These functions, i.e., kernels, can also present non linearity. Here, we apply the Gaussian radial basis function (RBF) kernel since it is more flexible than the linear one, and it has a better convergence time than the polynomial one.

3. EXPERIMENTAL SETUP AND RESULTS

This section describes the dataset that is used to validate the method for the classification of refreezing ice and the experimental setup for the feature extraction and the classification. The classification is performed considering all the features of the model and aims at labeling the pixels with $c_i, i = [1, ..., 4]$, representing the classes of layers, bedrock, refreezing ice, and EFZ and thermal noise.

We apply the method to two radargrams acquired in an airborne campaign in the north of Greenland in 2017 (see Figure 3). This campaign was chosen since it probes a region where there is significant evidence for the

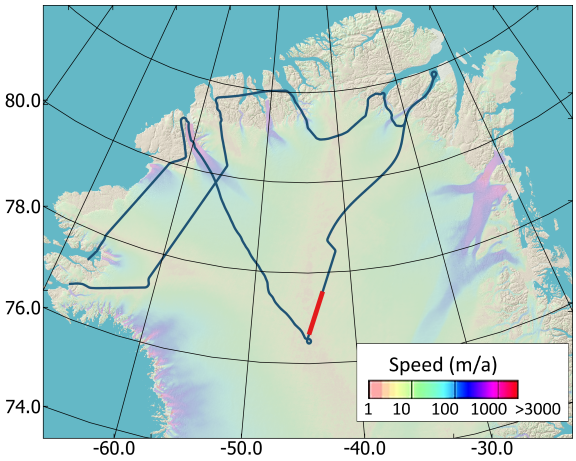


Figure 3: Ground track of the Campaign 20170413_01 in North Greenland. The position of the radargrams 45 and 46 is highlighted in red. The background shows the flowing velocity of glaciers.

formation of refreezing ice. Table 2 shows the parameters of the radar MCoRDS 15 that acquired the data and also the radargram characteristics, e.g., resolutions in the range and azimuth directions.

The data distributed by the CReSIS are already range compressed, azimuth focused via synthetic aperture radar (SAR) techniques, and the clutter is partially compensated. Further, we correct the data for the fluctuations due to the movements of the platform. While acquiring the data, an aircraft has three movements, i.e., pitch, roll, and yaw, that can influence and degrade the quality of the radargrams. Moreover, we concatenated the radargrams paying attention to remove the overlapping regions. Figure 5 shows the fluctuation corrected and concatenated radargram used for validating the proposed method.

A set of $J = J_c, c = [1, ..., C]$ samples representing the four classes were manually picked from the radargrams. These regions are selected in a way that there is no ambiguity regarding which target they belong to. To this end, we based the selection of these regions on examples of radargrams visually inspected and available in the literature 5,6. Hence, we labeled the samples and generated the training set T as a set of couples of sample and label covering all the classes.

For the feature extraction we apply a sliding window approach: the window is moved along the range and the azimuth directions independently. We chose a rectangular window since all the targets are elongated in the azimuth direction. Finally, the size of the window (n_w x n_h) is really important: the window should be big enough to be representative of the data. However, its size should be small enough to have locally representative feature and not penalizing the borders of the radargram. Considering this points, we chose a window of $n_w = 13$ and $n_h = 7$ pixels.

We choose the SVM classifier with the RBF kernel since it shows good performances in predicting radar data. To determine the parameters of the kernel, we apply a 10-fold cross-validation considering the range of the parameters $c \in [10^{-3}, 10^{8}]$ and $\gamma \in [10^{-2}, 10^{4}]$. From the 10-fold cross-validation, the optimal kernel parameters are obtained as $c = 256$ and $\gamma = 16$. Considering the set J of labeled samples, we uniformly extract J_l samples for the learning stage. The set J_l is divided in two sub-sets J_{cv} and J_{tr} that are used for the cross-validation and training phase, respectively. The remaining samples $J_{val} = J - J_l$ are used for the validation. To choose the value of J_l, we run experiments with different number of training samples to check the stability of the classifier. Here, we consider a training of 1%, 2%, and 5% of the reference set. For each of these training sets, we perform 10 experiments by changing the samples and we report here the average metrics results.

Table 3 shows the accuracy results for each class for the classifier trained with sets of different sizes. The overall accuracy for the three cases is comparable and has a value greater than 0.99. This proves that the geophysical model and the features defined in Section 2 characterize the properties of the refreezing ice and the other targets. Moreover, the accuracy results in Table 3 show that the method is robust, and not influenced by the number of samples in the training set.

Table 3: Accuracy results for the four classes considering different percentage of training set. For each training set, we performed 10 experiments and we report here the average of the accuracy results.

	Layers	**Refreezing Ice**	**Bedrock**	**EFZ and Thermal Noise**	**Overall Accuracy**
1%	0.9968	0.9821	0.9974	0.9945	0.9959
2%	0.9966	0.9871	0.9985	0.9920	0.9950
5%	0.9965	0.9884	0.9997	0.9920	0.9955
Dataset 1 7	0.9951	–	0.9751	0.9948	0.9909
Dataset 2 7	0.9853	–	0.9530	0.9834	0.9793

Table 4: Results considering the training set that considers 1% of the reference data.

	Layers	**Refreezing Ice**	**Bedrock**	**EFZ and Thermal Noise**
Sensitivity	0.9968	0.98207	0.99744	0.9945
Specificity	0.9994	0.9988	0.9996	0.9965
Precision	0.9998	0.96213	0.99573	0.9860
False Positive	0.0005	0.0012	0.0004	0.0034

Figure 4: Ground track in Greenland

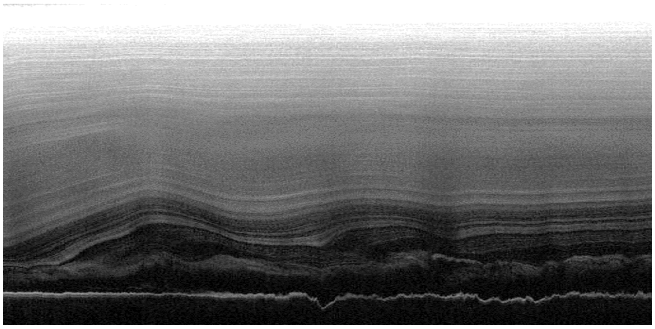


Figure 5: Radargrams 45 and 46 of the campaign 20170413_01 used as data set to prove the method. The radargrams are pre-porocessed (i.e., range compression, SAR focusing, and clutter reduction), corrected for the aircraft fluctuation, and concatenated.

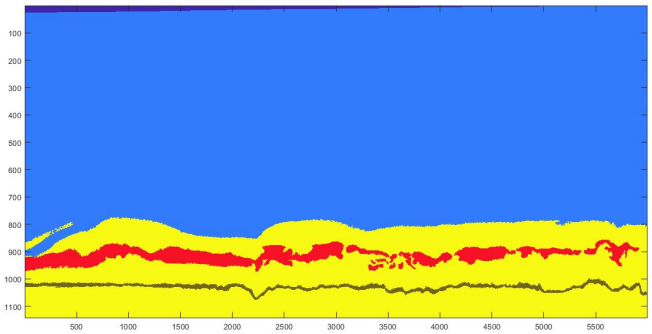
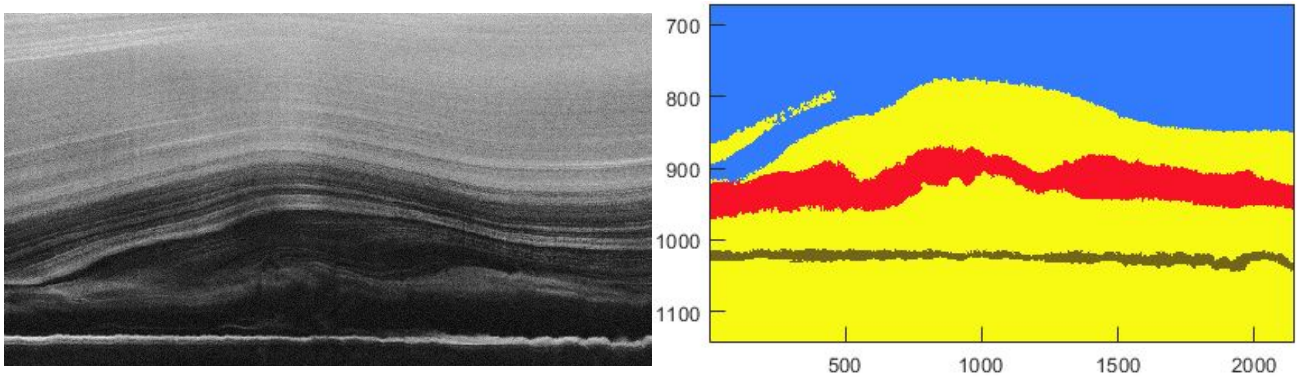


Figure 6: Classification map considering the training set that considers 1% of the reference data. The samples predicted as layers are in light blue, those classified as refreezing ice are in red, the yellow corresponds to samples classified as EFZ and thermal noise, and finally in brown the bedrock.



(a) Detail of the input data.

(b) Detail of the classification map.

Figure 7: Detail of the classification map with Figure (a) as input and 1% of the reference data for the training. Figure (b) is a detail of the classification map. The samples predicted as layers are in light blue, those classified as refreezing ice are in red, the yellow corresponds to samples classified as EFZ and thermal noise, and finally in brown the bedrock.

Figure 6 shows the classification map considering the radargrams in Figure 5 an the training set with 1% of the reference data. In the classification map, we can see that the four classes are well discriminated. The dark blue corresponds to free-space pixels, i.e., the pixels above the surface that are not taken into account in the classification. Samples predicted as layer are in light blue, those predicted as bedrock are in mustard color, and those as noise in yellow. The pixels predicted as refreezing ice are in green, and we can visually check that they correspond to the refreezing ice in the radargram. The overall accuracy of classification is $OA = 99.59\%$– Table 4 shows the sensitivity (i.e., true positive), specificity (i.e., true negative), precision (i.e., false negative), and false positive metrics. We can see the three classes are well discriminated– the sensitivity, specificity, and the precision have value greater that 0.99 for each class, while the false alarm is at maximum equal to 0.005. These results are consistent with the state-of-the-art ones 7.

Figure 7b shows the detail of the classification map obtained by considering as input the image in 7a. This consists of the initial 2500 traces and the last 400 samples of the radargram in Figure 5. Analyzing Figure 7b, we can see that there are no patches of refreezing ice in the layer region, and vice versa. The layers and the refreezing ice are clearly discriminated without any mis-classification, even though they have similar characteristics. However, some pixels in the lower part of the layers are classified as EFZ, i.e., in yellow. This is reasonable considering the detail in Figure 7a where a smooth passing from the layers to refreezing ice can be observed. The borders between classes are hard to classify since the properties of the signal change gradually from one target to the other.

4. CONCLUSIONS AND FUTURE WORK

This paper proposes an automatic method to extract information on basal the boundary conditions, and in particular on refreezing processes. These processes are related to the sliding movement of glaciers and the volume of ice melting at the basal interface. The information on the ice melting volume is crucial to better estimate the fresh water stored as ice, while the sliding movement should be studied to understand the melting in coastal areas in the framework of climate change.

We propose a geophysical and radiometric model for refreezing ice that describes its spatial, geological, statistical, and radiometric properties. From this model, we defined a set of feature able to discriminate reflections of refreezing ice from those of other structures in radargrams. Considering these feature, we classify the radargrams into four classes, i.e., layers,refreezing ice, echo-free zone and thermal noise, and bedrock. As classifier we chose the SVM since it allows an easy and fast validation of the proposed model and related feature. However, any classifier can be applied.

We tested our method on radargrams acquired by MCoRDS[15] in the North of Greenland. The results show the validity of the geophysical and radiometric model for refreezing ice. The accuracy for the refreezing ice class is higher than 90%, while the accuracy for the other classes is comparable with those in the literature.

As future work, we plan to perform an analysis on the sensitivity of the results by varying the size of the sliding window. Moreover, we want to validate the model and the method with other dataset acquired in other areas, e.g., in the Antarctica and Greenland. Further, we plan to model other structures related to the refreezing ice to better under basal processes and extract information on parameters related to the velocity of the sliding movement of the glacier.

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